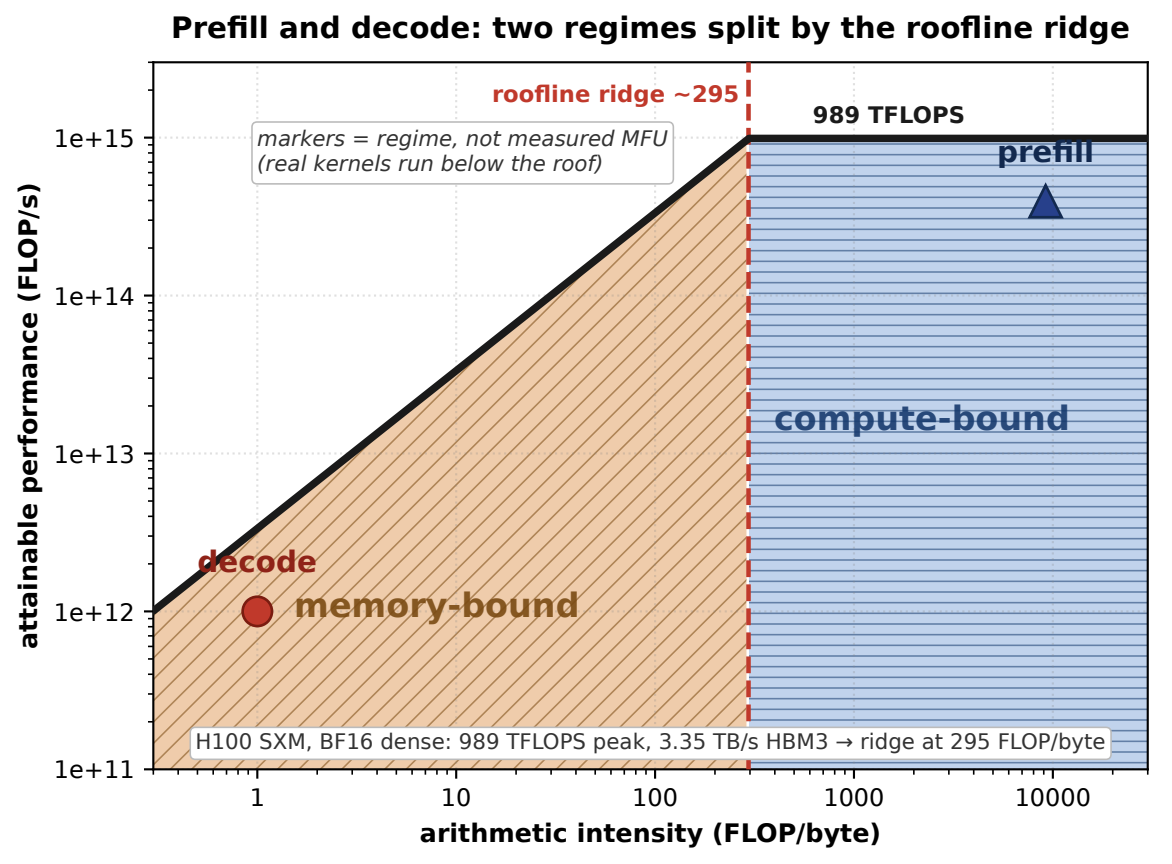
Prefill and decode: two regimes split by the roofline ridge
roofline ridge ~295
989 TFLOPS
markers = regime, not measured MFU
(real kernels run below the roof)
prefill
compute-bound
decode
memory-bound
H100 SXM, BF16 dense: 989 TFLOPS peak, 3.35 TB/s HBM3 → ridge at 295 FLOP/byte
attainable performance (FLOP/s)
1e+15
1e+14
1e+13
1e+12
1e+11
1
10
100
1000
10000
arithmetic intensity (FLOP/byte)